

5.4 Cameras

5.4.1 AiAF

Artificial Intelligence Auto Focus is Canon's patented focusing technology, which dynamically selects one or more focussing points based on factors like subject position and motion. It also detects camera position, as in, the focus automatically changes if the camera is held horizontally or vertically.

5.4.2 Aperture

The iris-like structure that controls the amount of light falling on the lens. The aperture is measured in f-stops. This range varies from $f/1.4$ to $f/22$, where f is the focal length of the lens.

5.4.3 CCD (Charge Coupled Device)

A CCD is a light-sensitive integrated circuit. It stores colour information for every pixel as electric charge, whose intensity depends upon the position of the colour in the colour spectrum. The charge for each pixel is read at one corner of the sensor and converted from analogue to digital. CCDs are used in digital cameras as sensors because they are extremely sensitive to colour variations and create low noise.

5.4.4 CMOS (Complementary Metal-Oxide Semiconductor)

CMOS is a technology that finds use in, among other things, camera sensors. It is the main competitor to CCD as a sensor technology. Though more susceptible to noise and less sensitive, they consume much less power, and are improving year after year in terms of quality.

5.4.5 Digid

Digital Image Core, or Digid, is a processor at the heart of any digital camera. Introduced by Canon, the Digid processor removes noise from the electric signals generated when light falls on the image sensor. It removes false colours, reproduces natural colours, and can deliver high-quality video and images even in adverse conditions like subject motion and poor light. The latest version is Digid III.